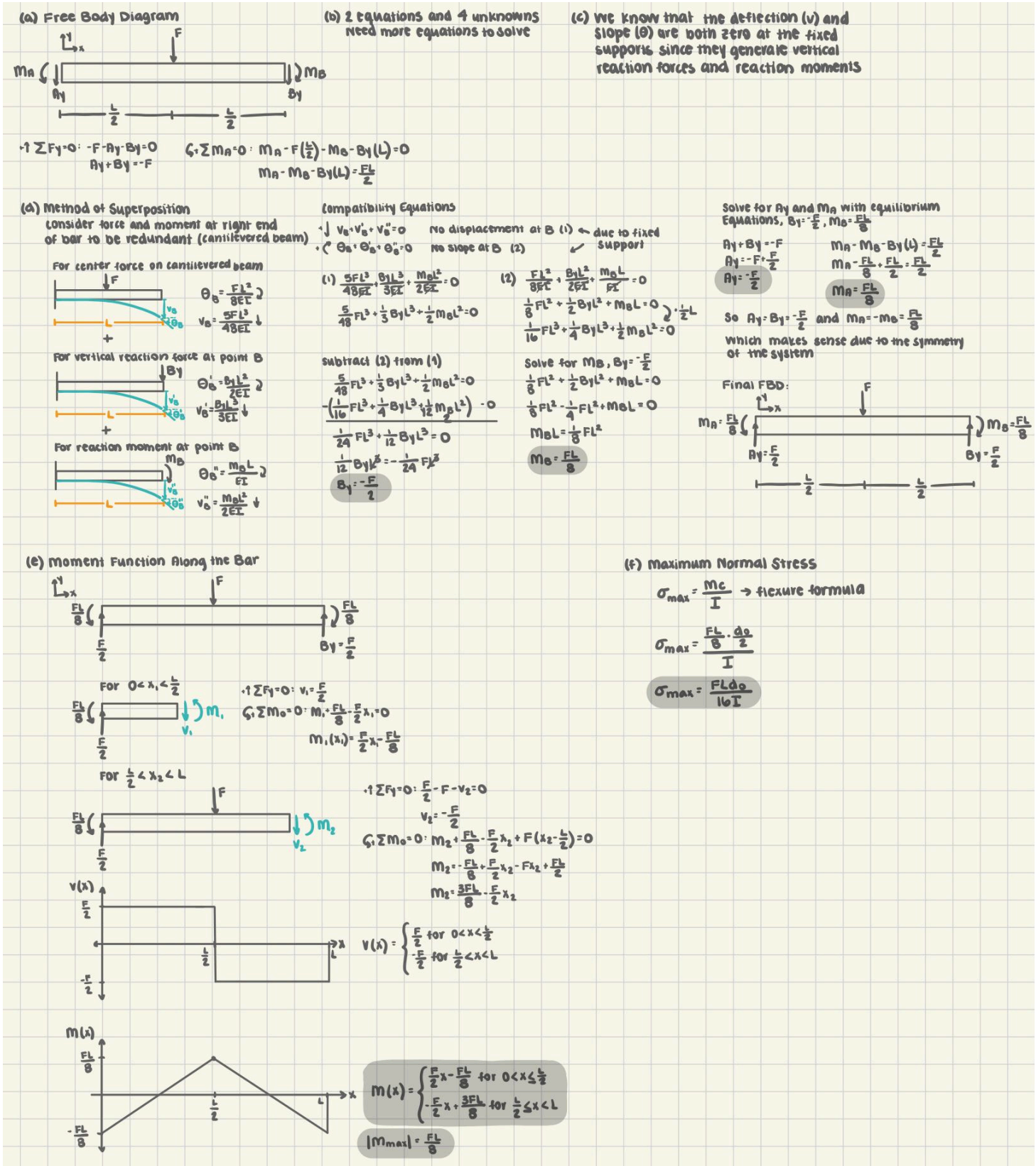
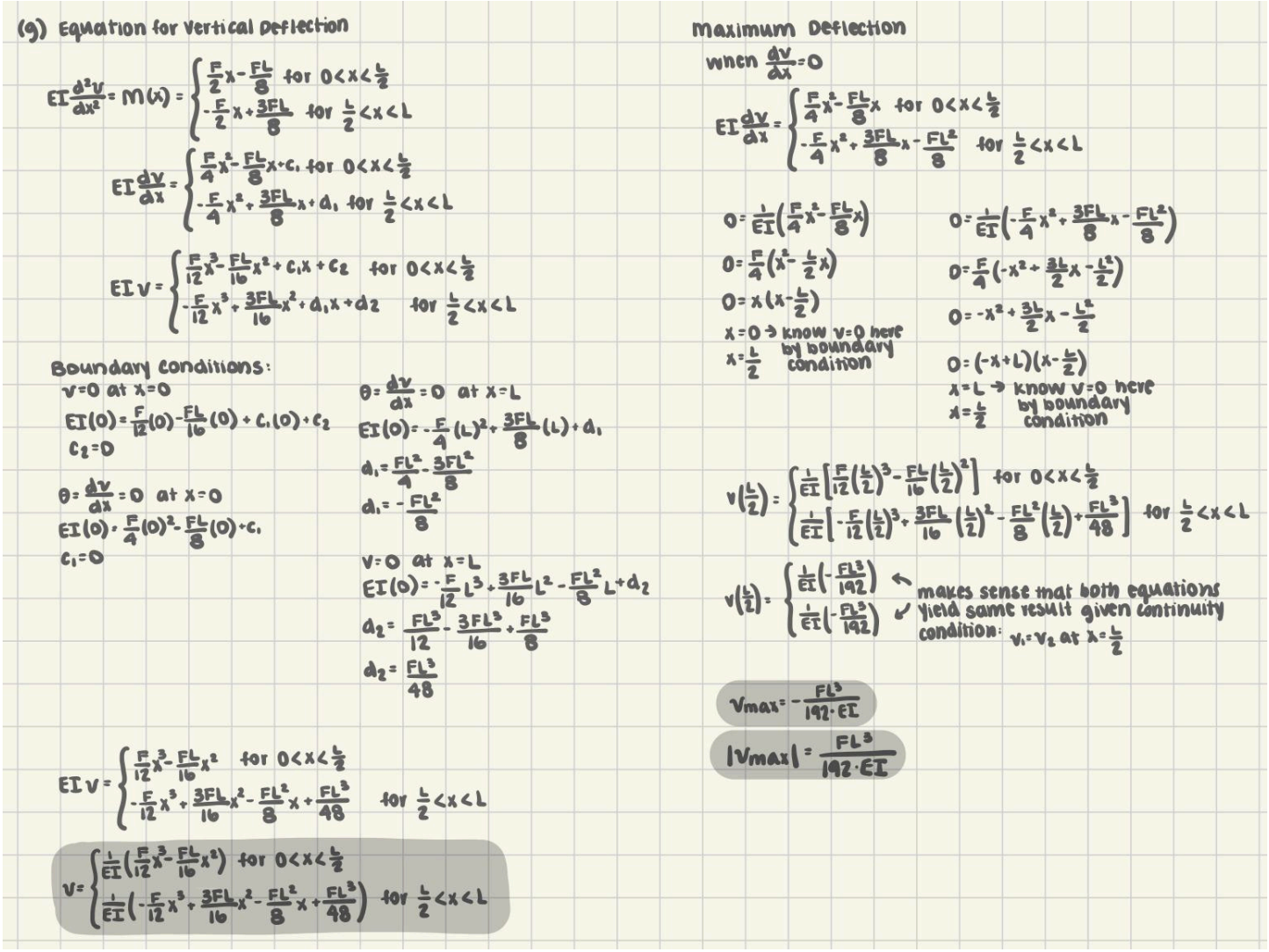
ME 20 Project 1

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Part 1: Determine the normal stress and deflection developed in the pull-up bar





Part 2: Comparing the peak normal stress and deflection with the ones from FEA (ME10)

F = 1300N; L = 770mm = 0.77m; d_0 = 25mm = 0.025m; d_i = 17mm = 0.017m; Iz = $\pi(d_0^4 - d_i^4)/64$;
E = 210GPa = 210E9 Pa

h) Find the numerical value of the maximum normal stress (from step f)

$$\sigma_{max} = \frac{FLd_0}{16I} = \frac{1300\ N * 770\ mm * 25\ mm}{16 * ((\pi * (25\ mm^4 - 17\ mm^4))/64)} = \ 103.7525\ MPa$$

i) Find the numerical value of the maximum vertical deflection (from step g)

$$\nu_{max} = \ -\frac{FL^3}{192 * EI} = \frac{1300\ N * 770\ mm^4}{192 * 210E3\ MPa * ((\pi * (25\ mm^4 - 17\ mm^4))/64)} = \ -0.976\ mm$$

j) Compare the values found in steps h and i to the ones you find in deliverables 1 through 6 in your ME10 assignment using the table below:

	From steps h & i	Deliverables 1 & 2	Deliverables 3 & 4	Deliverables 5 & 6
Maximum normal stress (MPa)	**103.7525 MPa**	**88.8393 MPa**	**88.8393 MPa**	**89.9554 MPa**
Maximum deflection (mm)	**-0.976 mm**	**-0.875 mm**	**-0.875 mm**	**-0.90625 mm**

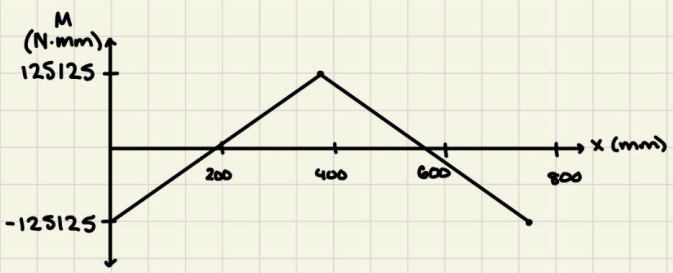
k) Comment on the sources of differences you observed in step j.

In comparison to the theoretical calculations, the max normal stress and deflection of the bar is higher than the FEA results. This makes sense as the loading conditions between the theoretical calculations and FEA are different, as the theoretical calculations have a concentrated point load exerting force directly on the middle of the bar, while the FEA has a distributed load. This distributed load means that the force is exerted over a greater surface area, meaning that the stress, or force per unit area will be lower for any given point, thus the maximum will be lower. Additionally, this holds for the deflection, as well, as it makes sense that the deflection would be lower for a distributed load as the stress is lower.

The last simulation in FEA is slightly more accurate than deliverables 1-4, as the fixed geometry simulation does not provide an accurate depiction of the stress and deflection the bar will experience. It holds that deliverables 1-4 will report the same maximum stress and deflection as the symmetry should simulate the same results between the full and half bars both with fixed geometry. The virtual wall has slightly higher max normal stress and deflection, as the bolts are less supportive than the fixed geometry, which supports the bar along the entire face of the wall support, thus the bar will be under more stress. Thus, deliverables 5&6 are closer to the theoretical calculations, but still, if one is to assume the load will be distributed over an actual pull-up bar, the theoretical calculations overestimate the maximum normal stress and deflection.

l) Redraw the moment diagram with the given numerical values and compare the shape of it to the stress plot you get from deliverable 1 (FEA). What do you notice?

l)

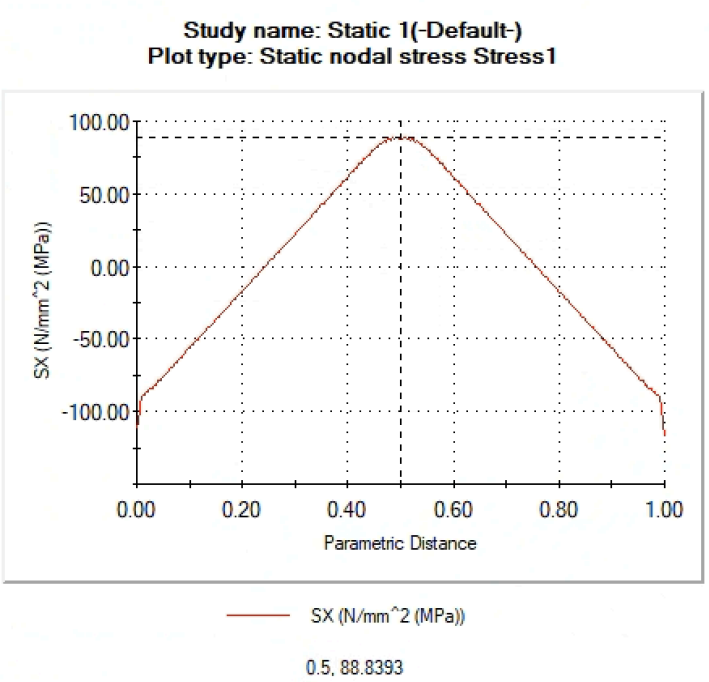


$$M(x) = \begin{cases} \frac{F}{2}x - \frac{FL}{8} & \text{for } 0 \le x \le \frac{L}{2} \\ -\frac{F}{2}x + \frac{3FL}{8} & \text{for } \frac{L}{2} \le x \le L \end{cases}$$

* plug in $F = 1300N$, $L = 770\,mm$

$$M(x) = \begin{cases} \frac{1300N}{2}x - \frac{(1300N)(770mm)}{8} & \text{for } 0 \le x \le \frac{770mm}{2} \\ -\frac{1300N}{2}x + \frac{3(1300N)(770mm)}{8} & \text{for } \frac{770mm}{2} \le x \le 770\,mm \end{cases}$$

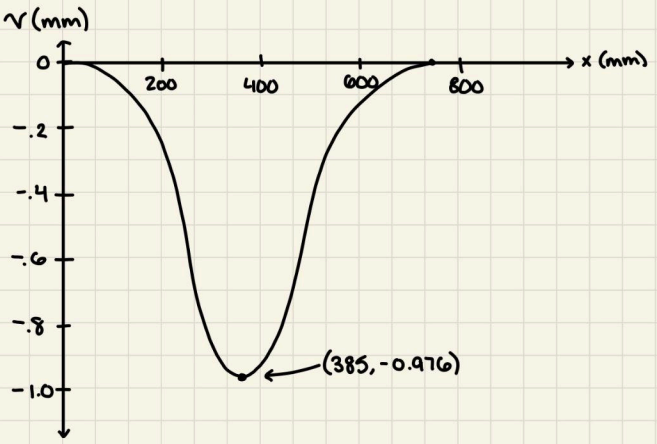
$$M(x) = \begin{cases} 650x - 125125 & \text{for } 0 \le x \le 385 \\ -650x + 375375 & \text{for } 385 \le x \le 770 \end{cases}$$



The shape of the theoretical moment diagram is similar to the stress plot from the FEA as they both have the peak at the center of the bar, which makes sense as the moment is proportional to the stress. However, for the FEA plot, there is a curved peak stress, which is due to the distributed load. In the theoretical calculations, since the force is concentrated at one point, the max stress will also be concentrated at that point, versus distributed over the area in the FEA, which makes the maximum stress peak in a curved manner as the beam will face the maximum stress throughout the portion over which the distributed load is applied.

m) Draw a schematic of the vertical deflection along the beam (using the equation you derived in step g) and compare it to the deflection curve from deliverable 2 (FEA). What do you notice?

m)



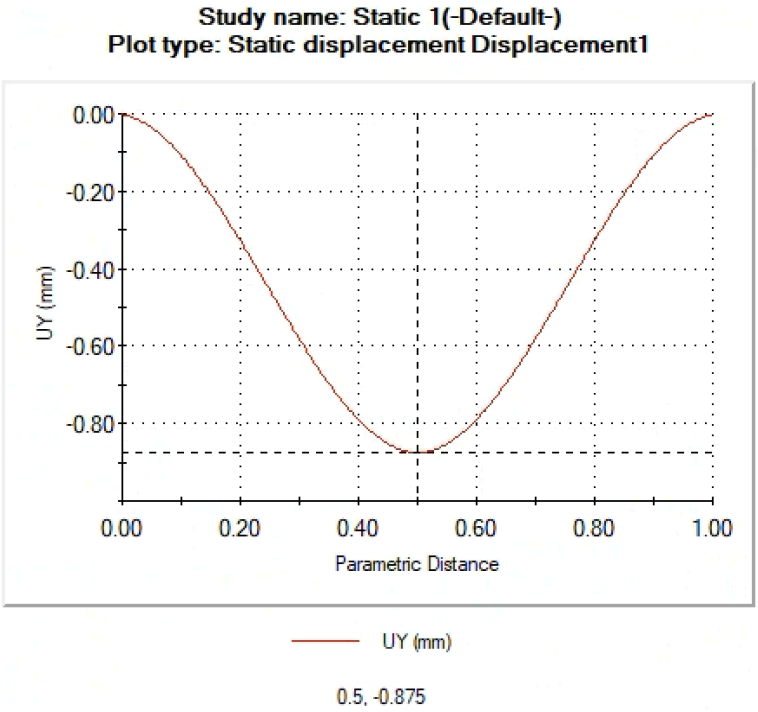
$$v = \begin{cases} \frac{1}{EI}\left(\frac{F}{12}x^3 - \frac{FL}{16}x^2\right) & \text{for } 0 \le x \le \frac{L}{2} \\ \frac{1}{EI}\left(-\frac{F}{12}x^3 + \frac{3FL}{16}x^2 - \frac{FL^2}{8}x + \frac{FL^3}{48}\right) & \text{for } \frac{L}{2} \le x \le L \end{cases}$$

* plug in $F = 1300N$, $L = .770m$, $E = 210GPa = 210E9\ Pa$

$$I = \frac{\pi\left[(.025m)^4 - (.017m)^4\right]}{64} = 1.5075E\text{-}8\ m^4$$

$$v = \begin{cases} 3.159E\text{-}4\left(108.33x^3 - 62.5625x^2\right) & \text{for } 0 \le x \le .385 \\ 3.159E\text{-}4\left(-108.33x^3 + 187.6875x^2 - 96.346x + 12.36\right) & \text{for } .385 \le x \le .770 \end{cases}$$

* Converted back into mm on graph



The shape of the vertical deflection graph is the same for the theoretical calculations and the FEA. The maximum deflection in the theoretical calculations is greater in magnitude than that of the FEA, which makes sense as stated in part k. The shape is parabolic due to the cubic functions from the theoretical displacement calculations. As a note, we converted the units for the deflection from mm and GPa to m and Pa, in order to keep all of the units consistent in the piecewise function, and we then converted the values we got for various points from meters to millimeters to plot them on the graph.

Conclusion:

In this project, we learned to apply the principle of superposition to determine the reaction forces of a pull-up bar when there are too few equilibrium equations to solve for all the unknown forces. By superimposing the individual forces acting on the bar and determining the individual deflections and slopes they inflict (from the known values for cantilevered beams), we could use the boundary conditions (that the slope and deflection are zero at the fixed supports) as compatibility conditions to obtain two more equations and solve for the unknown forces. We also applied the knowledge we learned in class about deriving the equation for the deflection by taking the second integral of the moment equation and using the boundary conditions to solve for unknown constants resulting from the integration. In problems solved in class and for the homework however, we often needed to use the continuity conditions to solve for these constants, however, we learned through this project that if the boundary conditions provide enough equations to solve for these constants, the continuity conditions are not needed. In this case, because both supports are fixed, the boundary conditions state that both slope and deflection are zero at these points (whereas for a pin, slope would not be zero). Thus, these four boundary conditions (two for each fixed support) provided the information necessary to solve for the four constants (two for each half of the bar) produced in the integration process.

Additionally, we learned how to utilize our skills in CAD to compare and contrast with theoretical calculations. Using SolidWorks to simulate the load on the pull-up bar was very helpful in visualizing the stress and deflection that the bar would face, and allowed us to compare the results from the study in order to verify our theoretical calculations were accurate. It was also very interesting to see how the various simulations we created compared to the theoretical point load, and how the distributed load and change between fixed geometry and a virtual wall with bolts affected the magnitude of the maximum stress and deflection. We learned that there are advantages and disadvantages to both approaches, as the theoretical calculations account for the worst-case scenario of the load, but might overestimate the stress and displacement that the actual pull-up bar will face, while the FEA provides a good visualization of an actual distributed load on the pull-up bar, but depending on the fixed wall conditions might underestimate the stress and displacement.

Overall, this project was an apt culmination of our theoretical approaches to calculating the happenings of force on systems and exploring how something that seems fairly simple like load on a pull-up bar has much more happening within than meets the eye. We were able to take the calculations and superposition tactics we learned in class to analyze the complexities of force on cantilever beams and move toward a real-life application of the effect of load on a pull-up bar and see using SolidWorks the intricacies of stress, strain, and resulting displacement. While we faced challenges in determining whether our calculations and simulations made sense at times, it was good to practice using our various resources for problem-solving and weighing the accuracies of each to determine that our calculations, FEA, and graphing paint a fairly accurate picture of the effects of a load placed on the center of a pull-up bar.